

Heartwatch Executive Summary

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Editor:

Grading Categories:

1. Critical thinking & analysis
2. Use of evidence/research
3. Development & structure
4. Language & style conventions

Atrial fibrillation (AFib) is a heart condition that causes an irregular rapid heartbeat. A 2019 study published by the NIH showed that 20% of AFib patients lacked confidence in obtaining and interpreting results. Ignoring AFib can lead to complications such as stroke and heart failure.

HeartWatch is a smartwatch app designed to address these challenges by integrating real-time AFib monitoring. Compatible with existing smartwatch technology, HeartWatch collects heart rate and Electrocardiogram (ECG) data and presents the information through easy-to-read graphs. By analyzing trends and individual health histories, HeartWatch provides users with clear and comprehensible health insights.

HeartWatch improves Medline Health by addressing its reliance on general “healthy” vital ranges (60-100 beats per minute), which overlook conditions like AFib (100-200 beats per minute). Instead, HeartWatch explains irregular vitals specifically for AFib and links them to health conditions, offering more accurate, actionable insights.

HeartWatch will meet FDA Class II medical device standards for moderate-risk devices, ensuring safety and effectiveness. With a development cost of \$243,985 over 40 weeks and six months for FDA clearance, the app will securely analyze heart rate and ECG data while complying with HIPAA regulations. Encryption and user education will mitigate security risks.

HeartWatch empowers patients with AFib to confidently monitor their heart health through clear, accessible insights. By simplifying vital data and providing actionable information, HeartWatch helps users take proactive steps to manage their condition and reduce the risk of complications like stroke and heart failure.